

Smart City EV Charging & Parking Management

1. Problem Statement

The Smart City EV Charging & Parking Management System is a database system designed to manage electric vehicle (EV) charging stations, parking spaces, reservations, charging sessions, payments, and dynamic pricing in an urban environment.

The system allows customers/drivers to register their electric vehicles and search for available parking slots and charging stations. A driver can make a reservation for a specific parking slot for a defined period of time. The system must store the reservation start time and end time in order to calculate the parking duration and corresponding parking fee.

Each parking location may contain multiple parking slots, and some slots are equipped with EV charging stations. These charging slots are exclusively reserved for Electric Vehicles. If a non-electric vehicle occupies an EV charging slot, the system automatically applies a parking violation fee. A charging station provides charging services with a specific maximum power capacity and charging rate.

When an EV uses a charging station, the system records a charging session, including the start time, end time, energy consumed in kilowatt-hours (kWh), and the applicable price per kWh. Additionally, if an EV remains parked in a charging slot after its charging session is complete, or exceeds the allowed time limit or maximum kWh capacity, the system applies an idle parking fee for the excess duration.

The system also supports dynamic pricing. Parking fees and charging rates may vary depending on the time period. For example, prices during peak hours may be higher than prices during off-peak hours. Therefore, the system must maintain pricing rules with their effective time periods and rates.

The system must also prevent conflicting reservations, ensure that a parking slot cannot be reserved by multiple customers during overlapping periods, and calculate the total cost of a parking and charging transaction accurately.

The database should support the following major functions:

- Manage customers and their electric vehicles.
- Manage parking locations and parking slots.
- Manage EV charging stations.
- Allow customers to reserve parking slots.
- Track parking duration.
- Record charging sessions.
- Track energy consumption in kWh.

- Apply different charging rates according to time periods.
- Apply different parking rates according to peak/off-peak periods.
- Apply parking violation fees for non-electric vehicles occupying charging slots.
- Calculate idle fees for EVs that exceed charging time or kWh limits.
- Calculate total fees (parking, charging, idle, and violation).
- Record payments.
- Maintain historical reservation, charging, pricing, and payment data.

2. Business Rules *parking apply to electric device , if exceed time will calculate fees of parking slot

Rule ID	Category	Business Rule
BR1	Customer & Vehicle	Each customer must have a unique CustomerID.
BR2	Customer & Vehicle	A customer can own zero, one, or many electric vehicles.
BR3	Customer & Vehicle	Each electric vehicle belongs to exactly one customer.
BR4	Customer & Vehicle	Each EV must have a unique VehicleID and a unique license plate.
BR5	Customer & Vehicle	A vehicle can make multiple parking reservations over time.
BR6	Customer & Vehicle	The system must distinguish between vehicle types (e.g., Electric Vehicle vs. Internal Combustion Engine).
BR7	Parking Location	The system manages multiple parking locations.
BR8	Parking Location	Each parking location must have a unique LocationID.
BR9	Parking Location	A parking location contains one or many parking slots.
BR10	Parking Location	Each parking slot belongs to exactly one parking location.
BR11	Parking Location	Each parking slot has a unique SlotID within the system.
BR12	Parking Location	A parking slot has a status such as Available, Reserved, Occupied, or Maintenance.
BR13	Parking Location	Parking slots equipped with EV charging stations are strictly reserved for Electric Vehicles.
BR14	Charging Station	A parking slot may or may not have an EV charging station (0..1).
BR15	Charging Station	Each charging station belongs to exactly one parking slot.
BR16	Charging Station	A charging station has ChargingStationID, ConnectorType, MaximumPowerKW, and Status.
BR17	Charging Station	A charging station can support multiple charging sessions over time.
BR18	Charging Station	A charging session belongs to exactly one vehicle and one charging station.
BR19	Reservation	A customer can create multiple reservations.
BR20	Reservation	Each reservation is made for exactly one vehicle.
BR21	Reservation	Each reservation reserves exactly one parking slot.
BR22	Reservation	Each reservation must contain ReservationID, StartDateTime, EndDateTime, and ReservationStatus.

BR23	Reservation	EndDateTime must be later than StartDateTime.
BR24	Reservation	Reservation duration is calculated as EndDateTime - StartDateTime.
BR25	Reservation	A parking slot cannot have two active reservations whose time intervals overlap.
BR26	Charging Session	A vehicle can have many charging sessions.
BR27	Charging Session	Each charging session belongs to exactly one vehicle.
BR28	Charging Session	Each charging session uses exactly one charging station.
BR29	Charging Session	A charging session records SessionID, StartDateTime, EndDateTime, and EnergyConsumedKWh.
BR30	Charging Session	EndDateTime must be greater than StartDateTime.
BR31	Charging Session	Energy consumption must be greater than or equal to zero.
BR32	Charging Session	Charging duration is calculated as EndDateTime - StartDateTime.
BR33	Charging Session	$\text{ChargingFee} = \text{EnergyConsumedKWh} \times \text{RatePerKWh}$.
BR34	Charging Session	If an EV remains plugged into a charging station beyond a specified grace period after the charging session ends (or exceeds a maximum kWh limit), an idle parking fee is calculated.
BR35	Dynamic Pricing	The system maintains charging-rate records with RateID, RatePerKWh, StartTime, EndTime, EffectiveFrom, EffectiveTo, and RateType.
BR36	Dynamic Pricing	A rate is valid only during its defined effective period.
BR37	Dynamic Pricing	Different rates may apply at different times.
BR38	Dynamic Pricing	If a charging session crosses multiple pricing periods, the system calculates the applicable charge for each period.
BR39	Parking Pricing	Parking fees may depend on the time period.
BR40	Parking Pricing	Parking rates can be defined as Peak, Normal, or Off-Peak.
BR41	Parking Pricing	The system stores the effective period of each parking rate.
BR42	Parking Pricing	Parking fee can be calculated based on parking duration and the applicable rate.
BR43	Parking Pricing	If parking duration crosses multiple pricing periods, the system calculates the fee for each applicable period.
BR44	Parking Pricing	If a non-electric vehicle (gasoline/ICE) occupies a charging slot, a specific parking violation fee is automatically applied.
BR45	Parking Pricing	The idle parking fee and violation fee are calculated based on the exceeded duration multiplied by a dynamic penalty rate.
BR46	Payment	A customer may make a payment for a completed parking/charging transaction.
BR47	Payment	Each payment has a unique PaymentID.
BR48	Payment	A payment records PaymentID, PaymentDateTime, Amount, PaymentMethod, and PaymentStatus.
BR49	Payment	$\text{TotalFee} = \text{ParkingFee} + \text{ChargingFee}$.
BR50	Payment	A payment can have statuses such as Pending, Paid, Failed, or Refunded.

3. ER/EER Diagram



